Experiment 2

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1 Title

Study of gamma-ray absorption in matter and inverse square law by gamma-ray source Na^{22}.

2 Aim

Our aim of this experiment is :

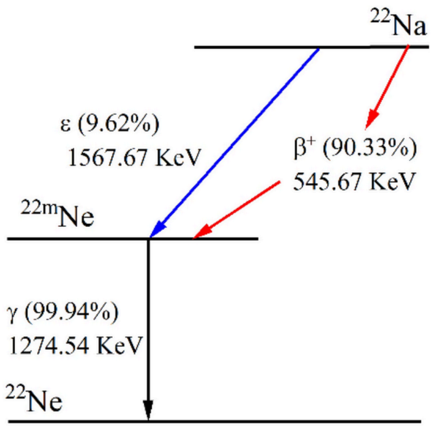
- To determine attenuation constant and half-value thickness of aluminium by keeping the distance of source and detector intact and putting aluminium blocks of different thicknesses in between them.
- To verify the inverse square law of radiation by varying the source-detector distance in the absence of any absorber.

3 Theory

3.1 Decay Scheme of $_{11}Na^{22}$:

Radioactive sodium-22 emits β^{+} and γ rays throughout its decay.

A proton in ^{22}Na converts into a neutron, emitting a positron and a neutrino, and leaves the daughter ^{22}Ne nucleus in an excited state.



$$^{22}\mathrm{Na} \xrightarrow[90.3\%]{\beta^{+},\, E_{\max}=0.545\ \mathrm{MeV}} {}^{22}\mathrm{Ne}^{*} + e^{+} + \nu_e$$

The nucleus captures an inner-shell electron, turning a proton into a neutron and emitting only a neutrino, again producing excited ^{22}Ne.

$$^{22}\mathrm{Na} \xrightarrow[9.7\%]{EC} {}^{22}\mathrm{Ne}^{*} + \nu_e$$

The excited ^{22}Ne nucleus de-excites by emitting a 1.275 MeV γ-ray, becoming stable ^{22}Ne.

$$^{22}\text{Ne}^* \rightarrow {}^{22}\text{Ne} + \gamma(1.275\ \text{MeV})$$

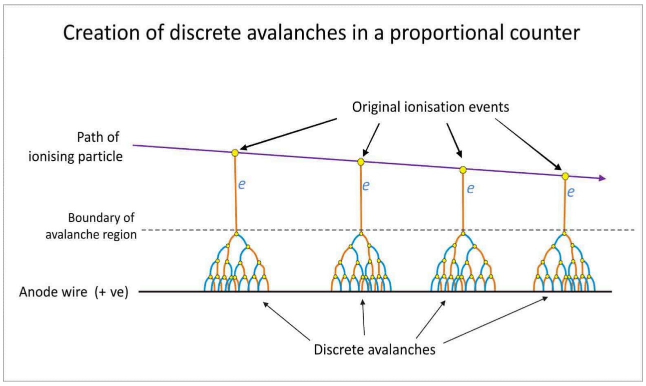
3.2 Geiger Miller (GM) Counter :

The detection of β & γ particles was carried out using a Geiger–Müller (GM) counter, which converts the ionizing radiation into measurable electrical pulses. It consists...

3.2.1 GM Tube :

The GM tube is a self-extinguishing halogen counting tube for the detection of α, β and γ radiation. It consists of a cylindrical cathode and a central anode wire filled with a low-pressure inert gas.

Townsend Avalanche:
When a radiation is passed through the GM tube, it knocks an electron from the outer orbit of that inert gas which further hits another molecule, ionising it. This creates an avalanche under the influence of a high electric field. It's called Townsend avalanche which produces a large current pulse.



Bremsstrahlung Effect :
When a stray electron enters the potential field of a nucleus, it slows down and emits an UV ray, which further can start an avalanche.

3.2.2 Counting Circuit :

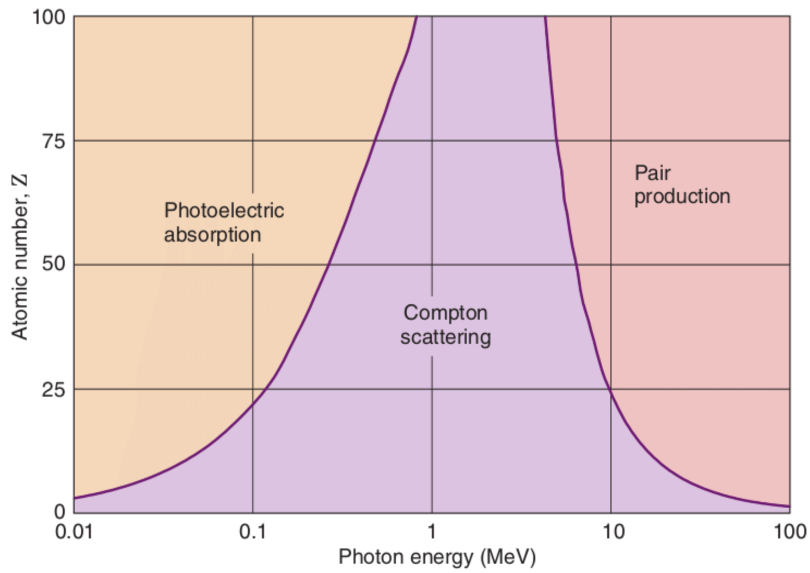
Each avalanche pulse is counted as a single event by the scaler–timer circuit.The apparatus is able to store the counts of pulses. Of each incoming pulse, the counter converts it into square wave, stores it in memory.

3.2.3 Dead Time of GM Tube :

As a full avalanche occurs throughout the tube, most of the molecules deliver their electrons to the central anode and form a barrier-like structure of anions around it. As they need some time to travel to cathode to get electrons, the GM tube becomes unresponsive to radiations, unable to initiate an effective avalanche. This time of inactivity is called the Dead time. It can vary from microseconds to milliseconds based on the size of the GM tube.

3.3 Absorption of γ ray :

3.3.1 Gamma Interactions : High energy γ photons can interact with matter mainly in 3 ways.



$fig: Z - E\ Diagram\ of\ photon\ interactions$

A. Photoelectric Effect :

Complete absorption of photon resulting ejection of an electron.

$$E_{kin} = h\nu - E_B$$

The process only takes place with K or L-electrons and occurs more often with substances with a high atomic number (Z). When the photon energy is too high, a nucleus with a high Z cannot handle the surplus of impulse either,that is why the photoelectric effect only occurs up to a limited energy value.

B. Compton Scattering :

Photon partially transfers its energy to electron, changing the photon's direction.

$$E_{\gamma}^{'} = \frac{E_{\gamma}}{1 + \frac{E_{\gamma}}{mc^2}(1 - cos\theta)}$$

C. Pair Production :

Only if γ-ray energy is above 1.02M eV ,photon converts into electron and positron.

3.3.2 Mass Attenuation : Whether γ-ray gets absorbed or scattered in the processes mentioned above, they don't make it to the detector. This can be characterized by a fixed probability of occurrence per unit path length in the absorber. The sum of these probabilities is simply the probability per unit path length that the gamma-ray photon is removed from the beam:

$$\mu = \tau(\text{photoelectric}) + \sigma(\text{Compton}) + \kappa(\text{pair})$$

This μ is called **linear attenuation coefficient**. If the number of photon transmitted without absorber is I_0 and number of photons after introducing a slab of thickness t is I. Then, the relation is:

$$\frac{I}{I_0} = e^{-\mu t}$$

The linear attenuation coefficient varies with the density (ρ) of the material. So generally we use **mass attenuation coefficient (κ)** , which is defined by:

$$\kappa = \frac{\mu}{\rho}$$

So, putting the value of κ into our previous attenuation formula, we get it in terms of mass attenuation coefficient:

$$\frac{I}{I_0} = e^{-\kappa\rho t}$$

This gives us,

$$ln(I) = ln(I_0) - \kappa\rho t$$

3.3.3 Half value thickness : Half value thickness is the thickness of the absorber at which the value of the incident intensity attenuated to half of its initial value. So putting $I = \frac{I_0}{2}$ & $t = t_h$

$$ln(I_0) - ln(2) = ln(I_0) - \kappa\rho t_h$$

$$\Rightarrow \quad t_h = \frac{ln(2)}{\kappa\rho}$$

3.4 Inverse square Law :

As the power is spread out in an area of $4\pi r^2$, the intensity I of the radiation received (which are equivalent to counts here) becomes inversely proportional to the square of the distance.

$$I \propto \frac{1}{r^2}$$

If we take the proportionality constant as A, then:

$$I = \frac{A}{r^2}$$

Now, if we take log on both side,

$$ln(I) = ln(A) - 2ln(r)$$

4 Data

4.1 Thickness vs Counts :

The screw gauge's least count on the linear scale was 0.5mm and for the circular scale, it was 0.01mm. We found out the zero error to be **-0.18mm** so we took each plate, measured their thickness and corrected them.

Aluminium plate no.	Thickness measurement via screw gauge			
	main scale reading (mm)	vernier scale (mm/100)	thickness (mm)	corrected thickness (mm)
1	0.5	30	0.8	0.98
2	0.5	30	0.8	0.98
3	0.5	30	0.8	0.98
4	0.5	30	0.8	0.98
5	0.5	30	0.8	0.98
6	5	46	5.46	5.64
7	10	18	10.18	10.36
8	10	18	10.18	10.36
9	10	18	10.18	10.36
10	10	18	10.18	10.36

Next, we measured the background radiation counts (in the absence of our source) and the count rate came out to be **0.3 counts/s.**
Then we took various different combinations of the plates and measured counts for 50 seconds for each thickness and corrected the count rate by subtracting background noise counts.

Plate Thickness (mm)	Counts				Count rate (s^{-1})	
	Count 1	Count2	Count3	Avg Count	Calculated rate	Corrected rate
0.98	32	25	27	28	0.56	0.26
1.96	23	28	28	26.33333333	0.5266666667	0.2266666667
2.94	31	38	30	33	0.66	0.36
4.9	22	27	28	25.66666667	0.5133333333	0.2133333333
5.64	30	24	29	27.66666667	0.5533333333	0.2533333333
7.6	24	28	27	26.33333333	0.5266666667	0.2266666667
10.36	28	27	25	26.66666667	0.5333333333	0.2333333333

10.54	26	26	23	25	0.5	0.2
15.26	19	28	25	24	0.48	0.18
16	24	30	31	28.33333333	0.5666666667	0.2666666667
20.72	27	23	26	25.33333333	0.5066666667	0.2066666667
20.9	26	26	22	24.66666667	0.4933333333	0.1933333333
23.66	26	24	26	25.33333333	0.5066666667	0.2066666667
26.36	24	28	19	23.66666667	0.4733333333	0.1733333333
31.08	17	27	21	21.66666667	0.4333333333	0.1333333333
31.26	15	33	18	22	0.44	0.14
34.02	23	31	31	28.33333333	0.5666666667	0.2666666667
36.72	19	19	22	20	0.4	0.1
41.44	24	26	14	21.33333333	0.4266666667	0.1266666667
44.38	22	18	15	18.33333333	0.3666666667	0.06666666667
47.08	16	15	21	17.33333333	0.3466666667	0.04666666667
51.98	20	20	16	18.66666667	0.3733333333	0.07333333333

4.2 Distance vs Counts : On the second day, we again measured the background noise counts for 600s and the counts are 149, 168 making the background radiation count rate to be **0.264 counts/s.**

Now, without any aluminium slab, we started taking counts measurements with varied distances.

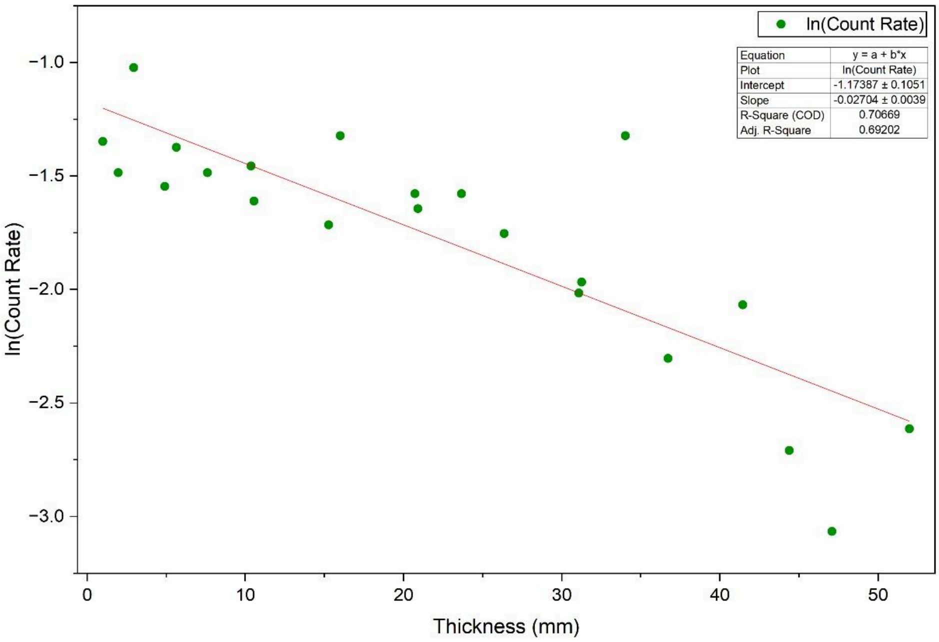
Distance (cm)	Counts			Time (s)	Count rate (s^{-1})	
	Count 1	Count 2	Avg count		Measured rate	Corrected rate
1.5	3056	3014	3035	60	50.58333333	50.31933333
2	1988	1991	1989.5	60	33.15833333	32.89433333
2.5	1371	1357	1364	60	22.73333333	22.46933333
3	945	967	956	60	15.93333333	15.66933333
3.5	722	754	738	60	12.3	12.036

4	547	588	567.5	60	9.458333333	9.194333333
4.5	437	413	425	60	7.083333333	6.819333333
5	318	376	347	60	5.783333333	5.519333333
5.5	302	304	303	60	5.05	4.786
6	236	248	242	60	4.033333333	3.769333333
6.5	208	215	211.5	60	3.525	3.261
7	187	167	177	60	2.95	2.686
7.5	164	172	168	60	2.8	2.536
8	150	135	142.5	60	2.375	2.111
8.5	132	127	129.5	60	2.158333333	1.894333333
9	102	127	114.5	60	1.908333333	1.644333333
9.5	108	89	98.5	60	1.641666667	1.377666667
10	95	103	99	60	1.65	1.386
10.5	88	85	86.5	60	1.441666667	1.177666667

5 Analysis

5.1 Determining Mass Attenuation Coefficient (κ) :

By fitting out plot with $y = a + bx$



$plot: \ fitted\ curve\ of\ Thickness\ vs\ log\ of\ count\ rate$

We got , $\ln(I_o)$ = a = -1.17387

linear attenuation coefficient μ = κρ = -b = 0.02704 ± 0.0039 mm^{-1}

As we know, density of aluminium ($\rho_{aluminium}$) = 2.7 g/cm^3 = 2.7 mg/mm^3

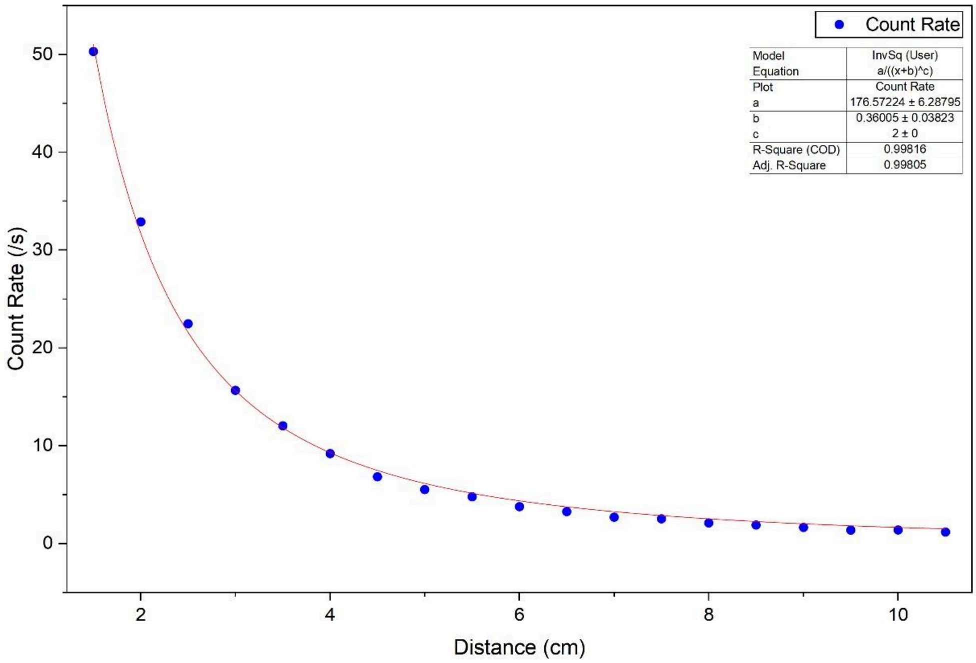
Mass attenuation coefficient :

$$\kappa = \frac{-b}{\rho} = \frac{0.02704}{2.7} = 1.0014 \times 10^{-2}\, mm^2/mg$$

half value thickness $t_h = \frac{ln\ 2}{\mu}$ = 25.634 mm

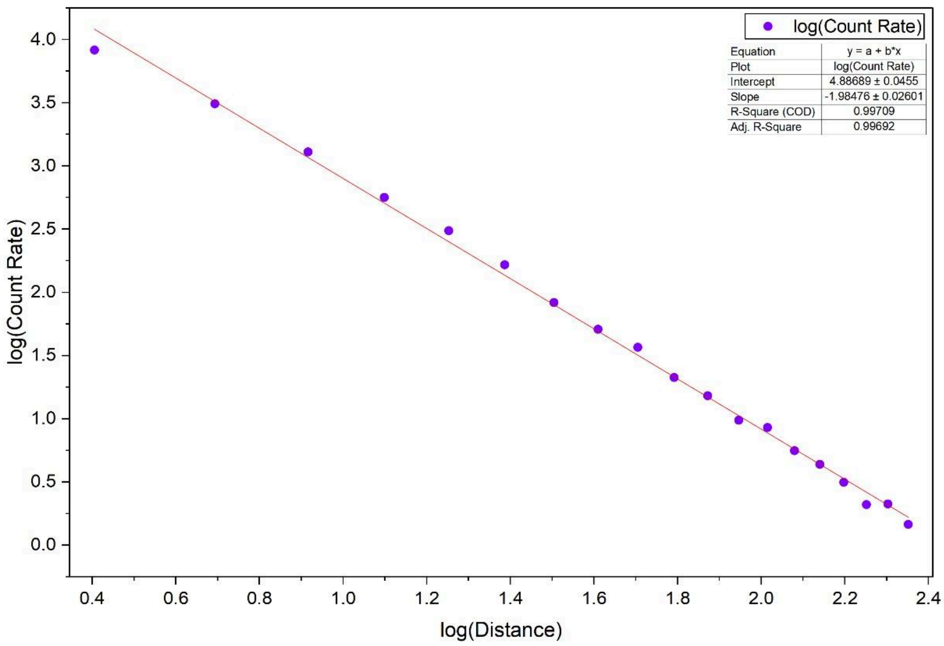
5.2 Proving Inverse Square Law :

By plotting our data and fitting it with $y = \frac{a}{(x+b)^2}$ we got:



$plot: \ fitted \ curve \ of \ distance \ vs \ count \ rate$

When we plotted both the natural log of distance and count rate, and fitted with $y = a + bx$,



$plot: \ fitted \ curve \ of \ log \ of \ distance \ vs \ log \ of \ count \ rate$

we got ,
Slope = -1.98476 which is very close to our desired value -2.
This graph shows the validity of inverse square law.

6 Sources of Error

We got errors in our fitted result. Also there were errors in measurements, the possible causes are,

6.1 Background Interference: Even if we've subtracted the background radiation counts, the ongoing experiments and existence of other sources in lab may cause uneven radiation leakage throughout the whole experiment which can hinder our result.

6.2 Instrumental errors: The base on which the distance measurements were marked may have errors. Also the screw gauge can have some errors.

6.3 Plate Alignment errors: While combined, the plates have some air gap in-between, which can cause extra scattering. Also the rubber bands weren't strong enough to hold the plates rigidly. So due to their small misalignments, errors may occur.

7 Conclusion

We completed our experiment with Sodium-22 as source of gamma ray and found out mass attenuation coefficient of aluminium to be $1.0014 \times 10^{-2} mm^2/mg$. The half value thickness of aluminium block was 25.634 mm. Also, we verified the inverse square law by our experiment with very small error.